

MATH 6241 – Stochastic Processes

❖ Queueing Theory

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- **Lindley's Equation**
- **Applications**
- **M/M/1**
- **M/M/c/K**

Lindley's Equation

We consider a **single-server queue** with customers indexed by arrival order.

- W_n^Q : **waiting time in queue** (excluding service) for the n th customer
- S_n : **service time** of the n th customer
- W_n : **time in system** for the n th customer

$$W_n = W_n^Q + S_n$$

- T_n : **arrival time** of the n th customer
- $X_n = T_n - T_{n-1}$: **interarrival time** between customers $n - 1$ and n th arrivals

□ Lindley's Recursion for Waiting Time in Queue

Theorem: The waiting time in queue evolves according to **Lindley's equation**:

$$W_{n+1}^Q = \max(0, W_n^Q + S_n - X_{n+1}), \quad W_1^Q = 0$$

Interpretation :

- The quantity $W_n^Q + S_n$ is the **total remaining workload** in the system just after customer n arrives.

During the interarrival time X_{n+1} , the server works continuously. Therefore:

- If $X_{n+1} \geq W_n^Q + S_n$, the system empties before customer $n + 1$ arrives, and customer $n + 1$ waits **zero** time.
- If $X_{n+1} < W_n^Q + S_n$, the system is still busy, and customer $n + 1$ waits exactly

$$W_n^Q + S_n - X_{n+1}$$

Equivalently, we write **Lindley's equation**

$$W_{n+1}^Q = \begin{cases} W_n^Q + S_n - X_{n+1} & \text{if } T_{n+1} < T_n + W_n^Q + S_n \\ 0 & \text{if } T_{n+1} \geq T_n + W_n^Q + S_n \end{cases}$$

Lindley's equation can be viewed as a **random walk**:

$$W_{n+1}^Q = \max(0, W_n^Q + \underbrace{S_n - X_{n+1}}_{\text{increments}})$$

The process evolves like a random walk but is **reflected at zero**.

□ Applications of Lindley's Recursions

We can use it to derive update formulas for other quantities of interest:

1. Time in System

Since, $W_n = W_n^Q + S_n$, we can derive a recursion for W_{n+1} :

$$\begin{aligned} W_{n+1} &= \max(0, W_n^Q + S_n - X_{n+1}) + S_{n+1} \\ &= \max(S_{n+1}, W_n^Q + S_n + S_{n+1} - X_{n+1}) \\ &= \max(S_{n+1}, W_n + S_{n+1} - X_{n+1}). \end{aligned}$$

This shows how the total time in system evolves from one customer to the next.

2. Departure Times

Define T_n^d = departure time of the n th customer.

Since departure occurs after arrival and completion of service,

$$T_n^d = T_n + W_n$$

Using $T_n = T_{n-1} + X_n$, we obtain

$$T_n^d = T_{n-1} + X_n + W_n.$$

Taking differences yields the recursion

$$T_{n+1}^d = T_n^d + T_{n-1} + X_n + W_n.$$

This formula is particularly useful for **event-driven simulations**, where arrival and departure times are generated sequentially.

3. Counting Processes

Consider time-indexed system quantities

$$N(t) = \text{number of arrivals in } [0, t] = \max\{n: T_n \leq t\}$$

$$N^d(t) = \text{number of departures in } [0, t] = \max\{n: T_n^d \leq t\}.$$

$$L(t) = \text{number of customers in the system at time } t.$$

$$L^Q(t) = \text{number of customers waiting in queue at time } t.$$

Then,

$$L(t) = \max(0, N(t) - N^d(t))$$

in a **single-server system**, at most **one customer can be in service** at any time.

$$L^Q(t) = \max(0, N(t) - N^d(t) + 1)$$

Lindley's equation and the associated recursions:

- require **no assumptions** on independence or distributions,
- hold **pathwise**, for every realization of arrivals and services,
- form the backbone of **stochastic simulations**,
- allow analysis of **stability** and long-run behavior.

Lindley's equation shows that queueing systems behave like a **reflected random walk**:

$$Workload_{n+1} = \max(0, Workload_n + Input - Output).$$

This viewpoint connects queueing theory to:

- renewal theory,
- random walks,
- heavy-traffic limits,
- and reflected Brownian motion.

➤ **Algorithm:** Single-Server Queue Simulation (Lindley-Based)

- Inputs**
- Interarrival times: $\{X_n\}_{n=1}^N$
 - Service times: $\{S_n\}_{n=1}^N$
 - Number of customers to simulate: N

Initialization

$$T_0 = 0, T_0^d = 0$$

$$W_1^Q = 0$$

$$T_1 = X_1$$

$$W_1 = S_1$$

$$T_1^d = T_1 + W_1$$

Recursive Update (for $n = 1, 2, \dots, N - 1$)

Step 1: Arrival time

$$T_{n+1} = T_n + X_{n+1}$$

Step 2: Queue waiting time (Lindley's equation)

$$W_{n+1}^Q = \max(0, W_n^Q + S_n - X_{n+1})$$

Step 3: Time in system

$$W_{n+1} = W_{n+1}^Q + S_{n+1}$$

Step 4: Departure time

$$T_{n+1}^d = T_{n+1} + W_{n+1}$$

State Variables at Time t

For any observation time t :

Arrival count

$$N(t) = \max\{n: T_n \leq t\}$$

Departure count

$$N^d(t) = \max\{n: T_n^d \leq t\}$$

Number in system

$$L(t) = \max(0, N(t) - N^d(t))$$

Queue length

$$L^Q(t) = \max(0, N(t) - N^d(t) - 1)$$

Outputs

- Waiting times: $\{W_n^Q\}$
- System times: $\{W_n\}$
- Departure times: $\{T_n^d\}$
- Sample paths of $L(t)$ and $L^Q(t)$
- Time averages (after burn-in):

$$\bar{W}, \bar{W}^Q, \bar{L}, \bar{L}^Q$$

□ The $M/M/1$ Queue

Workhorse model: the $M/M/1$ queue. It is a special case of the single-server FIFO queue.

Assume

- The interarrival times $\{X_n\}$ are IID exponential with mean $1/\lambda$.
- The service times $\{S_n\}$ are IID exponential with mean $1/\mu$.
- The two sequences are independent.

Then:

- The arrival process $N(t)$ is a **Poisson process** with rate λ .
- Because both arrival and service times are memoryless, the number of customers in the system evolves as a **continuous-time Markov chain**.
- This Markov chain is a **birth–death process**.

State Description

Let

$L(t)$ = number of items in the system at time t ,

$L^Q(t)$ = number of items in the queue at time t .

The states of the Markov chain

$$\{0, 1, 2, 3, \dots\}$$

where state n means there are n customers in the system.

The transitions are:

- **Births (arrivals):** $n \rightarrow n + 1$ at rate λ .
- **Deaths (departures):** $n \rightarrow n - 1$ at rate μ for $n \geq 1$.

The chain has the standard birth–death structure.

$$0 \xrightleftharpoons[\mu]{\lambda} 1 \xrightleftharpoons[\mu]{\lambda} 2 \xrightleftharpoons[\mu]{\lambda} 3 \xrightleftharpoons[\mu]{\lambda} \dots$$

Stationary Distribution

Define the stationary probabilities:

$$p_n = \lim_{t \rightarrow \infty} P(L(t) = n), \quad n = 0, 1, 2, \dots$$

In steady state, flow into each state equals flow out. The **rate-balance equations** are:

$$\lambda p_n = \mu p_{n+1}, \quad n \geq 0.$$

Equivalently,

$$p_{n+1} = \frac{\lambda}{\mu} p_n.$$

Define the traffic intensity

$$\rho = \frac{\lambda}{\mu}.$$

Then by recursion,

$$p_n = \rho^n p_0.$$

To find p_0 , use normalization:

$$1 = \sum_{n=0}^{\infty} p_n = p_0 \sum_{n=0}^{\infty} \rho^n.$$

The geometric series converges only if $\rho < 1$, giving

$$\sum_{n=0}^{\infty} \rho^n = \frac{1}{1 - \rho}.$$

Hence,

$$p_0 = 1 - \rho.$$

Therefore, the **stationary** distribution is

$$p_n = (1 - \rho)\rho^n, n \geq 0$$

This is a **geometric distribution**.

Lemma

Let $\rho = \lambda/\mu$.

- If $\rho < 1$, then the chain is stable and

$$P(L(t) = n) \rightarrow p_n = (1 - \rho)\rho^n \text{ as } t \rightarrow \infty.$$

- If $\rho \geq 1$, then no stationary distribution exists and

$$P(L(t) = n) \rightarrow 0 \text{ for every fixed } n.$$

The queue grows without bound.

We can use these expressions to compute average quantities when $\rho < 1$

- **Mean number in the system**

$$E[L] = \ell = \sum_{n=0}^{\infty} n p_n = (1 - \rho) \sum_{n=0}^{\infty} n \rho^n = \frac{(1 - \rho)\rho}{(1 - \rho)^2} = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda}$$

- **Mean Time in the System (Little's Law)**

By Little's Law, $l = \lambda w$. Therefore,

$$w = \frac{l}{\lambda} = \frac{1}{\mu - \lambda}$$

- **Mean Waiting Time in Queue**

Since total time in system equals waiting plus service,

$$w = w^Q + E[S] = w^Q + \frac{1}{\mu}$$

Therefore,

$$w^Q = w - \frac{1}{\mu} = \frac{\rho}{\mu - \lambda}$$

- **Mean Queue Length**

Apply Little's Law to the queue, $l^Q = \lambda w^Q$. Thus,

$$l^Q = \lambda \frac{\rho}{\mu - \lambda} = \frac{\rho^2}{1 - \rho}$$

Example.

Machines break down according to a Poisson process with rate

$\lambda = 6$ per hour and are repaired with exponential service times at rate

$\mu = 8$ per hour. The cost of lost production is \$10 per hour for each broken machine.

What is the average cost per hour due to broken machines?

First compute the traffic intensity: $\rho = \frac{\lambda}{\mu}$

For an $M/M/1$ queue, the mean number of customers in the system is

$$l = \frac{\rho}{1 - \rho}$$

Each broken machine costs \$10 per hour, hence

$$\text{Average cost per hour} = 10 \times l = 10 \times 3 = 30.$$

We can also compute the **distribution** in the $M/M/1$ queue

- W : the *time spent in the system* (waiting + service), and
- W^Q : the *time spent waiting in the queue* (before service begins),

$$\mathbb{P}(W > t) = e^{-(\mu-\lambda)t}, \quad t \geq 0.$$

W is an exponential random variable with rate $\mu - \lambda$.

$$\mathbb{P}(W^Q > t) = \rho e^{-(\mu-\lambda)t}, \quad t \geq 0,$$

$$\mathbb{P}(W^Q = 0) = 1 - \rho.$$

W^Q has a **mixed distribution**:

$$W^Q = \begin{cases} 0, & \text{with probability } 1 - \rho, \\ \text{Exponential}(\mu - \lambda), & \text{with probability } \rho. \end{cases}$$

Recall Kendall's Notation

Naming convention for FIFO queue with IID interarrivals and IID service times:

A/S/c/K

- ***A*** is the arrival process (examples are M (Poisson process), D (deterministic) and G (general))
- ***S*** is the mathematical distribution of the service time (*e.g M (exponential) and G (general)*)
- ***c*** is the number of servers
- ***K*** is the capacity of the queue, omitted if unlimited

□ Example: $M/M/3/5$

This is a multi-server queue with **finite capacity**.

- Arrivals follow a Poisson process with rate λ .
- Each server has exponential service time with rate μ .
- There are **3 servers**.
- The **total system capacity is 5** customers (including those in service).
- If an arrival finds 5 customers already in the system, it is **blocked and lost**.

Hence the system states are

$$\{0, 1, 2, 3, 4, 5\}$$

where the state n means there are n customers in the system.

Because of the finite capacity, the system is **always stable**, for any value of λ and μ .

- **Arrival (birth):** $n \rightarrow n + 1$ at rate λ , $n = 0, 1, 2, 3, 4$, and arrivals are **blocked** when $n = 5$.
- **Service completion (death):**

$$n \rightarrow n - 1 \text{ at rate } \begin{cases} n\mu, & n = 1, 2, 3, \\ 3\mu, & n = 4, 5, \end{cases}$$

since at most 3 servers can work simultaneously.

Stationary Probabilities

Let $p_n = \lim_{t \rightarrow \infty} P(L(t) = n), \quad n = 0, 1, \dots, 5.$

The rate-balance equations are:

$$\begin{aligned}\lambda p_0 &= \mu p_1, \\ \lambda p_1 &= 2\mu p_2, \\ \lambda p_2 &= 3\mu p_3, \\ \lambda p_3 &= 3\mu p_4, \\ \lambda p_4 &= 3\mu p_5.\end{aligned}$$

From these we can express everything in terms of p_0 , and impose normalization:

$$1 = \sum_{n=0}^5 p_n.$$

Hence

$$p_0^{-1} = 1 + \rho + \frac{\rho^2}{2} + \frac{\rho^3}{6} + \frac{\rho^4}{18} + \frac{\rho^5}{54}$$

From these we can express everything in terms of p_0 :

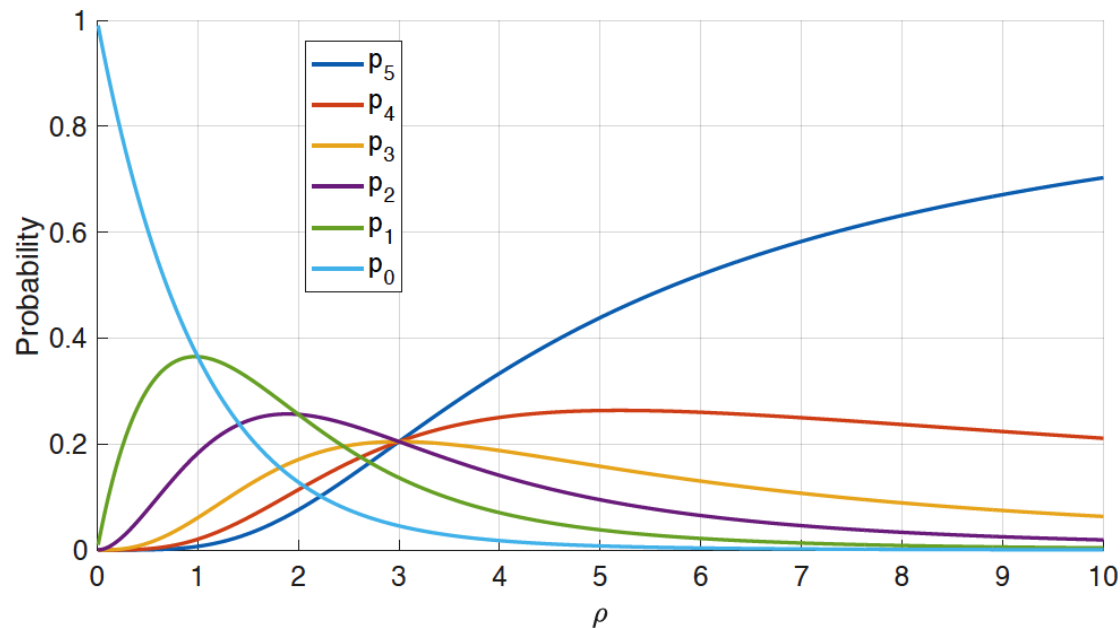
$$p_1 = \rho p_0,$$

$$p_2 = \frac{\rho^2}{2} p_0,$$

$$p_3 = \frac{\rho^3}{6} p_0,$$

$$p_4 = \frac{\rho^4}{18} p_0 = \left(\frac{\rho}{3}\right) \frac{\rho^3}{6} p_0,$$

$$p_5 = \frac{\rho^5}{54} p_0 = \left(\frac{\rho}{3}\right)^2 \frac{\rho^3}{6} p_0.$$



Probability the System Is Full

The probability that an arriving customer is blocked is exactly

$$p_5 = \left(\frac{\rho}{3}\right)^2 \frac{\rho^3}{6} p_0$$

This is the **blocking probability**.

Mean System Size and Queue Length

The mean number of customers in the system is

$$l = \mathbb{E}[L] = p_1 + 2p_2 + 3p_3 + 4p_4 + 5p_5$$

The mean number waiting in the queue is:

- When $n \leq 3$, nobody waits.
- When $n = 4$, one person waits.
- When $n = 5$, two people wait.

$$\ell^Q = p_4 + 2p_5$$

Effective Arrival Rate

Not all arrivals enter the system. Those that see state 5 are rejected.
Thus the **effective arrival rate** is

$$\lambda_{eff} = \lambda(1 - p_5)$$

This is the rate at which customers actually enter and get service.

Little's Law for a Finite-Capacity System

Little's Law must use λ_{eff} , not λ :

$$\ell = \lambda_{eff} w, \ell^Q = \lambda_{eff} w^Q,$$

where

- w = mean time in the system,
- w^Q = mean waiting time in the queue.

Thus

$$w = \frac{\ell}{\lambda_{eff}}, w^Q = \frac{\ell^Q}{\lambda_{eff}}$$

Numerical Example

Suppose:

- $\lambda = 8$ arrivals/hour,
- mean service time = 20 minutes = $1/3$ hour, so $\mu = 3$ per hour,
- Hence

$$\rho = \frac{\lambda}{\mu} = \frac{8}{3}.$$

Using the normalization formula:

$$p_0 = \frac{1}{1 + \rho + \frac{\rho^2}{2} + \frac{\rho^3}{6} + \frac{\rho^4}{18} + \frac{\rho^5}{54}} \approx 0.0637.$$

From this we compute:

$$\begin{aligned} \ell &= 2.74, \\ \ell^Q &= 0.497, \\ p_5 &= 0.159, \\ \lambda_{eff} &= \lambda(1 - p_5) = 8(1 - 0.159) = 6.727. \end{aligned}$$

Now apply Little's Law:

$$w = \frac{\ell}{\lambda_{eff}} = \frac{2.74}{6.727} \approx 0.407 \text{ hours},$$
$$w^Q = \frac{\ell^Q}{\lambda_{eff}} = \frac{0.497}{6.727} \approx 0.074 \text{ hours}.$$

That is

Mean time in system:

$$w \approx 0.407 \text{ hours} \approx 24.4 \text{ minutes}.$$

Mean waiting time in queue:

$$w^Q \approx 0.074 \text{ hours} \approx 4.4 \text{ minutes}.$$

□ Example: $M/M/1/K$

This example studies how to choose the service rate μ so as to **maximize long-run profit** in an $M/M/1/K$ queue.

- Arrivals: Poisson with rate λ .
- Service times: exponential with rate μ .
- Single server.
- System capacity: K customers (including the one in service).
- If an arrival finds K customers already present, it is rejected (lost).

Economic assumptions:

- It costs $c\mu$ dollars per hour to operate the server at rate μ .
- Each customer successfully served brings in A dollars.
- We want to choose μ to maximize profit in steady state.

Profit Rate

In stationarity, the **departure rate** equals the **effective arrival rate**, i.e., the rate at which customers actually enter the system. Denote it by λ_{eff} .

Then the profit per hour is

$$\Pi(\mu) = A \lambda_{eff} - c \mu$$

So profit = (revenue per customer $\times$ customers served per hour) – (cost of service capacity).

Case 1: $K = \infty$ (No capacity limit)

If $K = \infty$, no customers are ever blocked, so
 $\lambda_{eff} = \lambda$.

Thus,

$$\Pi(\mu) = A\lambda - c\mu.$$

For stability we require $\mu > \lambda$. Under this constraint, $\Pi(\mu)$ is decreasing in μ , so it is maximized by choosing μ as small as possible:

$$\mu \downarrow \lambda.$$

Then

$$\Pi_{\max} = (A - c)\lambda.$$

This corresponds to operating the server just barely fast enough to keep the system stable, at the cost of extremely long queues and delays.

So economically it is optimal, but operationally it is disastrous.

Case 2: $K = 1$

Now the system has only two states: $\{0,1\}$

State 0: system empty.

State 1: system full (one customer being served).

CTMC transitions:

$$0 \xrightarrow{\lambda} 1, \quad 1 \xrightarrow{\mu} 0.$$

The stationary probabilities satisfy

$$\lambda p_0 = \mu p_1, \quad p_0 + p_1 = 1.$$

Solving, $p_0 = \frac{\mu}{\lambda + \mu}, p_1 = \frac{\lambda}{\lambda + \mu}.$

An arrival is accepted only when the system is empty, so

$$\lambda_{eff} = \lambda(1 - p_1) = \lambda p_0 = \frac{\lambda \mu}{\lambda + \mu}.$$

Hence profit per hour is $\Pi(\mu) = A \frac{\lambda \mu}{\lambda + \mu} - c\mu$

Optimization for $K = 1$

Differentiate with respect to μ :

$$0 = \frac{d\Pi}{d\mu} = A \frac{\lambda^2}{(\lambda + \mu)^2} - c.$$

Solve for μ ,

$$\mu_{\max} = \lambda \left(\sqrt{\frac{A}{c}} - 1 \right)$$

This only makes sense when $A \geq c$; otherwise, revenue per customer is not worth the operating cost.

Case 3: $K = 2$

Now states are: $\{0,1,2\}$. Arrivals are blocked in state 2.

The stationary probabilities satisfy

$$p_1 = \rho p_0, \quad p_2 = \rho^2 p_0, \quad \rho = \frac{\lambda}{\mu},$$

and normalization gives

$$p_0 = \frac{1}{1 + \rho + \rho^2}.$$

The blocking probability is

$$p_2 = \frac{\rho^2}{1 + \rho + \rho^2}.$$

Hence the effective arrival rate is

$$\lambda_{eff} = \lambda(1 - p_2) = \lambda \frac{1 + \rho}{1 + \rho + \rho^2}.$$

Profit per hour becomes

$$\Pi(\mu) = A\lambda \frac{1 + \rho}{1 + \rho + \rho^2} - c\mu, \quad \rho = \frac{\lambda}{\mu}.$$

Setting $d\Pi/d\mu = 0$ leads to the equation

$$\frac{A}{c}(2 + \rho)\rho^3 = (\rho^2 + \rho + 1)^2, \quad \rho = \frac{\lambda}{\mu}.$$

Closed formula for ρ is huge. When $A/c \gg 1$, an approximation can be derived:

$$\mu_{\max} = \frac{\lambda}{\rho} \approx \lambda \left(\frac{2A}{c} \right)^{1/3}$$

❑ Erlang Loss System: $M/M/k/k$

This model is called an **Erlang loss system**. It has:

- Poisson arrivals with rate λ ,
- Exponential service times with rate μ ,
- k identical servers,
- **No waiting room.**

So:

- If fewer than k servers are busy, an arriving customer is immediately served.
- If all k servers are busy, the customer is **blocked and lost**.

There is no queue at all; customers either get service immediately or leave.

Stationary Distribution

Let $N(t)$ be the number of customers in service at time t .

The state space is $\{0, 1, 2, \dots, k\}$

This is a birth–death process with:

Birth rate:

$$\lambda_n = \lambda, \quad n = 0, 1, \dots, k-1,$$

and $\lambda_k = 0$ (blocking).

Death rate:

$$\mu_n = n\mu, \quad n = 1, 2, \dots, k.$$

The stationary probabilities are

$$p_n = \frac{\rho^n}{n!} \left(\sum_{j=0}^k \frac{\rho^j}{j!} \right)^{-1}, \quad n = 0, 1, \dots, k,$$

where $\rho = \frac{\lambda}{\mu}$.

Equivalently, since

$$\mu^{-1} = \mathbb{E}[S] \text{ (mean service time),}$$

we often write

$$a = \rho = \lambda \mathbb{E}[S],$$

called the **offered traffic** (measured in Erlangs).

Blocking Probability

The system is blocked exactly when all servers are busy, i.e., when $N(t) = k$.

Define

$$B(k, a) = \mathbb{P}(\text{system is blocked}) = p_k.$$

From the stationary distribution,

$$B(k, a) = \frac{a^k / k!}{\sum_{j=0}^k a^j / j!}$$

This is the famous **Erlang B formula**.

It gives the long-run fraction of arriving customers that are blocked.

- $a = \lambda \mathbb{E}[S]$ measures how much traffic is offered to the system.
- k measures how many servers are available.
- $B(k, a)$ decreases rapidly as k increases.
- This formula is fundamental in telecommunication, call centers, and capacity planning.

Example 1: Telephone Exchange

Calls arrive at rate

$$\lambda = 2 \text{ calls per minute.}$$

Each call lasts on average

$$\mathbb{E}[S] = 1 \text{ minute.}$$

The exchange has $k = 4$ lines.

So:

$$a = \lambda \mathbb{E}[S] = 2.$$

The fraction of blocked calls is

$$B(4,2) = \frac{\frac{2^4}{4!}}{\sum_{j=0}^4 \frac{2^j}{j!}} \approx 0.095$$

So about **9.5% of incoming calls are blocked.**

Example 2: Adding Two More Lines

Now suppose we increase capacity to $k = 6$, keeping the same traffic $a = 2$.

Then

$$B(6,2) = \frac{\frac{2^6}{6!}}{\sum_{j=0}^6 \frac{2^j}{j!}} \approx 0.012$$

So only about **1.2% of calls are blocked**

1. The Erlang loss model has no waiting: customers are either served immediately or lost.
2. The blocking probability depends only on:
 - the number of servers k ,
 - the offered traffic $a = \lambda \mathbb{E}[S]$.
3. Adding even a small number of extra servers can drastically reduce blocking.
4. This is why the Erlang B formula is still a cornerstone of teletraffic engineering and call-center design.